# Prerana Nale

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## Education

**Arizona State University** Tempe, AZ

*Master of Science in Software Engineering; GPA: 4.00/4.00* *Aug 2024 – May 2026*

## Skills

**AI & ML:** RAG Pipelines, LLMs, Retrieval & Vector Search, Model Evaluation, OpenAI API, LangGraph, ChromaDB, MLflow, Redis Streams, Prompt Engineering, SVD

**Languages & Frontend:** Python (Flask, FastAPI), JavaScript/TypeScript, Java (Spring Boot), Golang, React, Node.js

**Cloud & DevOps:** AWS (EC2, S3, RDS, IAM, KMS, Cognito), Docker, Terraform, Jenkins, GitHub Actions, CI/CD Pipelines

**Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQL Server, ChromaDB

**Process & Delivery:** Agile/Scrum, REST APIs, Microservices, TDD (pytest, JUnit, Vitest), Technical Documentation, Git

## Experience

**Software Engineer Intern** Jul 2026 – Present

*Ipser Lab LLC* *Remote, US*

- Built full-stack features with React, TypeScript, and Java REST APIs in Agile sprints; engineered reusable UI components to improve performance.
- Added automated unit and integration tests across microservices, integrated into the deployment pipeline.

**Instructional Assistant, Principles of Programming** Aug 2025 – Dec 2025

*Arizona State University* *Tempe, AZ*

- Led weekly Java labs for 150+ students on OOP and algorithms; co-authored grading rubrics and code-review feedback used by faculty.

**Software Engineer** Sep 2021 – Jan 2024

*Statics Solutions* *Pune, India*

- Built full-stack enterprise applications in Python, JavaScript, and Java (Spring Boot REST APIs + React) processing millions of records at 75% faster throughput; sustained 90%+ automated test coverage (pytest, JUnit).
- Rewrote schemas, added indexes, and profiled stored procedures on busy PostgreSQL and MySQL endpoints, cutting query latency by 40%.
- Decomposed a monolith into event-driven microservices secured with JWT and RBAC across 5 services, enabling API-level integration and independent, low-risk deployments.
- Automated releases with Jenkins and Docker CI/CD pipelines, cutting time-to-market by 60% (10-day manual deploys to 4-day automated releases); partnered with product and QA stakeholders across 8 Agile sprints.

## Projects

**Data Flywheel Pipeline – Self-Improving Recommendation System**

*FastAPI, Redis Streams, SVD, MLflow, React, Docker Compose, pytest*

- Architected an end-to-end ML platform across 7 independently deployable microservices (Inference, Feedback, Feature Pipeline, Scheduler, Trainer, Redis, MLflow) orchestrated in Docker Compose; feedback streamed via Redis Streams triggers autonomous retraining at 50+ events, eliminating every manual model-promotion step.
- Enforced automated evaluation gates (RMSE $\leq 0.947$, 100% catalog coverage) that block promotion on regression; matrix factorization model trained on 100k MovieLens ratings hot-swaps into production without downtime.

**MedCareBot – RAG Health Education Chatbot (Capstone)**

*React, Django, PostgreSQL, OpenAI GPT-4o-mini, ChromaDB, AWS Cognito, pytest, Vitest*

- Built a production RAG pipeline that embeds and indexes 83,843 FDA drug records in ChromaDB, retrieves relevant chunks at query time, and generates grounded answers with GPT-4o-mini; added LLM-generated knowledge checks to evaluate user comprehension.
- Delivered 136 automated tests (Vitest + pytest) across 5 Agile sprints in a 6-engineer team; integrated AWS Cognito auth and PostgreSQL-backed chat history, with the test suite catching 3 critical retrieval bugs before UAT.

**Self-Healing CI/CD Agent – Autonomous Test Repair**

*LangGraph, Groq LLM, Python*

- Built an autonomous LangGraph agent that detects failing tests, diagnoses root cause, patches code, and reruns CI unattended.

**ExceptionDesk – Delivery-Exception Resolution Agent (In Progress)**

*Python, Google ADK, OR-Tools, scikit-learn, Docker*

- Building a multi-agent system that retrieves policy (RAG), predicts delay risk, re-plans routes with an optimizer, and requires human approval for costly actions; evaluating it against a scripted baseline on held-out scenarios.

**Orange Sulphur – Accessible E-Commerce Platform**

*React 19, TypeScript, Redux Toolkit, Tailwind CSS, Web Speech API*

- Shipped a live storefront (React 19): 95/100 Lighthouse (top 2% globally), WCAG 2.1 AA, voice navigation, colorblind modes.